Technical Service Bulletin

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Incorrect Coolant Pre-Heater Installation Resulting in Piston Scuffing on QSK23 Engines	

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Core Issue

The coolant pre-heater has **not** been correctly connected to the engine for proper heating of the coolant. This results in the engine **not** being warmed properly prior to use at rated speed and load. This can lead to a scuffed piston(s).

Confirmation

Figures 1 through 4 below show center line piston scuffing that resulted from an engine **not** being warmed properly prior to use at rated speed and load.

Figure 1, Anti-Thrust Side Piston Scuff on Cylinder Liner

Figure 2, Thrust Side Piston Scuff on Cylinder Liner

Figure 3, Anti-Thrust Side Piston Scuff

Figure 4, Thrust Side Piston Scuff

Check the coolant pre-heater coolant inlet and outlet connections. The coolant pre-heater coolant inlet **must** be connected to the water inlet connection (2 or 3). The coolant pre-heater outlet **must** be connected to the cylinder block cover plate on the exhaust side of the engine (1). See Figure 5.

Note : The cylinder block cover plate may need to be modified to add a coolant pre-heater outlet connection location.

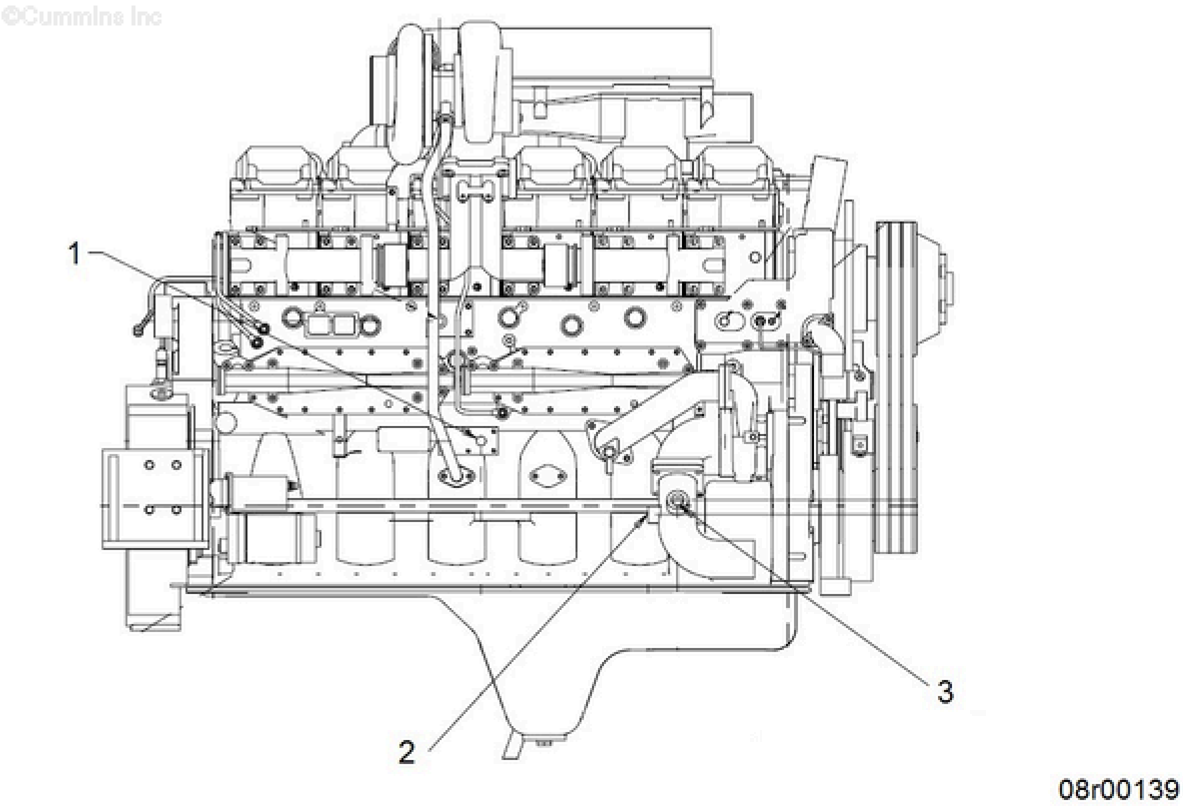

Figure 5, Coolant Pre-Heater Connection Ports

For coolant flow information, use the following procedure in the QSK23 Troubleshooting and Repair Manual, Bulletin 4021375. Refer to Procedure 200-003 in Section F. (/qs3/pubsys2/xml/en/procedures/89/89-200-003.html)

Resolution

If the coolant pre-heater coolant inlet and outlet connections are incorrect, correct them per Figure 5 above. This will ensure proper coolant pre-heater operation and heating of the engine.

In addition to a coolant pre-heater, there are other cold weather operating techniques to consider:

- Ensure the coolant temperature is 10°C [50°F] and the oil temperature is -4°C [25°F] before taking the engine off of low idle.
- Use the correct grade of lubricating oil for the current ambient conditions. Use the following procedure in the QSK23 Owners Manual, Bulletin 4915552. Refer to Procedure 018-003 in Section V. (/qs3/pubsys2/xml/en/procedures/102/102-018-003.html)
- For QSK23 used in Hitachi EX1200 excavators, the engine control module (ECM) calibration **must** be D50098 revision 02 or higher. This calibration will delay idle increase speed in cold ambient conditions.

For additional cold weather operation information. Refer to Service Bulletin, Operation of Diesel Engines in Cold Climates, Bulletin 3379009 (/qs3/pubsys2/xml/en/bulletin/3379009.html).

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

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2015-1-7	Module Created

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